ARUN SINGH

AI/ML Engineer | Machine Learning | Deep Learning | Computer Vision | Applied AI | IIT Jodhpur

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PROFILE SUMMARY

AI/ML-focused student pursuing a B.S. in Applied Artificial Intelligence and Data Science at IIT Jodhpur, with hands-on experience building machine learning, deep learning, computer vision, recommendation, NLP, and AI-assistant applications using Python. Experienced in data preprocessing, feature engineering, model training, evaluation, exploratory analysis, prompt engineering, and practical AI solution development. Seeking AI/ML Engineer, Machine Learning Engineer, or AI Engineering internship opportunities.

TECHNICAL SKILLS

Programming: Python, SQL

Machine Learning: Scikit-learn, Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Supervised & Unsupervised Learning, Recommendation Systems

Deep Learning: TensorFlow, Keras, PyTorch fundamentals, Neural Networks, CNN fundamentals

AI & NLP: Computer Vision, Object Detection Concepts, NLP Fundamentals, Sentiment Analysis, Prompt Engineering, AI Agents, LLM Applications

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Data Cleaning, Data Wrangling, EDA, Statistics & Probability

Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Power BI, Tableau

EDUCATION

Indian Institute of Technology (IIT) Jodhpur — B.S. in Applied Artificial Intelligence and Data Science | Ongoing

Relevant Coursework: Machine Learning, Deep Learning Fundamentals, Artificial Intelligence, Data Structures & Algorithms, Database Management Systems, Linear Algebra, Statistics & Probability

AI/ML PROJECTS

TrafficGuard-AI — AI-Based Traffic Violation Detection

Python | Computer Vision | Machine Learning | AI

- Developed an AI-based traffic monitoring solution designed to analyze city traffic-camera footage and identify potential traffic violations.
- Designed a workflow for AI-based violation detection and generation of digital challan information for identified vehicles.
- Applied computer-vision concepts to automate traffic monitoring and support more efficient traffic enforcement.

Movie Recommendation System

Python | Machine Learning | Recommendation Systems

- Built a movie recommendation application that recommends relevant movies based on user preferences and movie similarity.
- Applied data preprocessing and recommendation techniques to generate personalized results.
- Designed the project as a practical AI application focused on improving content discovery and user experience.

MindCheck — AI Mental Health Checkup

Python | AI | NLP | Data Analysis

- Developed an AI-based mental health checkup concept to help users assess their responses and receive structured insights.
- Applied AI/NLP-oriented logic to process user inputs and present understandable results.
- Focused on responsible, user-friendly interaction and clearly separating informational insights from professional medical advice.

JARVIES — AI Personal Assistant

Python | AI Agents | LLMs | Voice Interaction

- Built an AI assistant concept designed to understand natural-language and voice commands and respond conversationally.
- Designed command-processing workflows for task assistance, contextual understanding, and human-like interaction.
- Explored agent-style automation to connect user requests with practical computer and productivity actions.

AI Resume Shortlisting System

Python | NLP | Machine Learning | Resume Screening

- Built an AI-based resume shortlisting project to compare candidate resumes against job requirements.
- Used resume information and job-description requirements to support relevance scoring and candidate ranking.
- Designed the workflow to help recruiters reduce manual screening effort while keeping human review in the decision process.

Neural Network Digit Recognition

Python | TensorFlow | Keras | MNIST

- Built and trained a neural network to classify handwritten digits using the MNIST dataset.
- Experimented with network architecture and activation functions to understand their effect on classification performance.
- Implemented an end-to-end deep-learning workflow covering training and evaluation.

Student Performance Prediction

Python | Scikit-learn | Pandas | Machine Learning

- Cleaned and analyzed student academic data and engineered features for predictive modeling.
- Built a machine-learning model achieving 86% prediction accuracy.
- Used statistical analysis to identify factors associated with student performance and translate results into interpretable insights.

House Price Prediction

Python | Scikit-learn | Linear Regression

- Developed a regression model to estimate house prices using features including area, location, and room count.
- Evaluated model performance using R^2 and Mean Squared Error (MSE).
- Visualized predicted versus actual values to analyze model performance and identify improvement areas.

Iris Flower Classification

Python | Scikit-learn | K-Nearest Neighbors

- Implemented a K-Nearest Neighbors classifier for Iris flower classification.
- Tuned model parameters and compared performance across configurations to select an effective model.

COVID-19 Data Trends Analysis

Python | Pandas | Matplotlib

- Analyzed public COVID-19 datasets to identify infection trends across regions and time periods.
- Performed data cleaning, exploratory analysis, and visualization to communicate regional and temporal patterns.

CERTIFICATIONS & TECHNICAL ACTIVITIES

- Google Cloud Innovator Program Member
- Google Developer Program Member
- Google Cloud & NVIDIA Community Member
- Google Maps Platform Innovators Member
- Google Skills / Cloud Learning Badges
- Participation in AI/ML, cloud, developer, and technology-focused learning programs and communities

STRENGTHS & LANGUAGES

Strengths: Problem Solving • Analytical Thinking • Technical Learning • Data-Driven Decision Making • Teamwork • Communication

Languages: Hindi (Native) • English (Fluent)